

AttentionNGP: Semantic Lifting and Distillation

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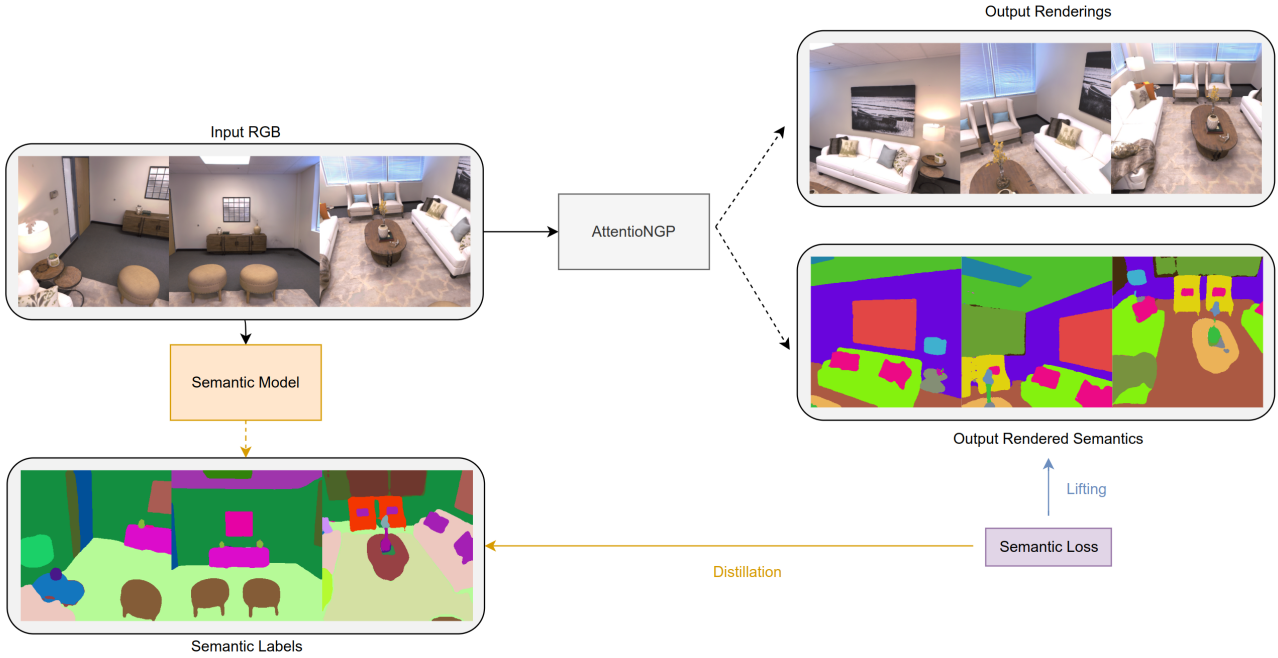


Figure 1. For any scene given by a set of RGB images, we machine generate the corresponding semantic labels using an off-the-shelf semantic segmentation model. Then, using our novel formulation of NeRF, AttentionNGP, we lift these paired RGB-semantic labels into a multiview-consistent 3D space. Once trained, the AttentionNGP model can be used to fine-tune the off-the-shelf semantic model by distilling its 3D-aware knowledge back through the model. Furthermore, by using the generalizable attention layers of our NeRF model we can theoretically transfer 3D aware knowledge without querying the underlying semantic model *at all*.

Abstract

When generalizing pre-trained 2D feature (in our case, semantic) field models to novel scenes and lighting conditions, typically performance is significantly degraded, and there are significant issues in multi-view consistency. Lifting these inconsistent labels into a implicit 3D scene representation has been proposed to alleviate this problem, but strong inconsistencies in the underlying feature fields can stop convergence in the implicit scene representation, especially when looking at the feature field components independently (pixel-wise). Instead, our proposed solution leverages the explicit stricture of Instant-NGP’s multi-resolution feature grid computes a local self-attention per query-ppoint to provide local

consistency and reasoning when lifting into 3D. Then, we distill the implicit scene representation’s global multi-view consistency into the pre-trained 2D feature field model as a fine-tuning step, improving the 2D model’s 3D aware performance on unseen data.

1. Introduction

High-level and complex 3D scene understanding is becoming increasingly important as computer vision is integrated into downstream applications such as self-driving, robotic-perception, security, AR and VR environments, etc. Fundamentally, scene understanding is the method by which raw signals from sensors observing the scene are transformed

into human-interpretable semantics that we can interpret and use for further control. Ideally, the scene understanding system should be able to generalize to novel scenarios out of the training distribution in order to adapt its operation when faced with new inputs. One avenue by which this task is achieved in the 2D case is through semantic segmentation and its variants (instance segmentation, panoptic segmentation).

Nonetheless, while there has been significant breakthroughs in 2D scene understanding through segmentation, especially as we get increasingly larger models, generalizability when it comes to novel scenes is severely hampered, especially when focusing on consistency across multiple views of the same scene with respect to occlusion and view-dependent effects, purely stemming from the lack of a 3D inductive bias within these models. An ideal 3D scene understanding model would be structured entirely in 3D, taking away the costly (in compute and performance) translation step between the known 2D supervision and required 3D models. However, such a model is extremely challenging even considering the required hardware components, and as such our next step is to improve the aforementioned translation step.

There are many recent works [17] [15] [11] [8] that attempt to tackle this translation step on various domains and applications ranging from continual adaptation to panoptic segmentation. However, the core conceit in many of these papers is that the 3D semantic field is learned *dependent* on position, making the learned semantics a property of the space rather than a consequence of the underlying feature/density field. This issue dampens the ability of a neural representation to generalize semantic understanding to new scenes based on local embeddings, which can be used to supervise the underlying 2D semantic models on completely novel scenes (with no semantic ground-truth) with a strong 3D prior.

In this work, we introduce a method by novel scenes with model generated semantic labels can be lifted into a 3D semantic field through neural rendering (Fig. 1). Importantly, we augment this *lifting* process with a self-attention component that is used to generalize and robustify the conversion process between the classical geometric field that underlies the NeRF into the semantic field, which allows for semantic field generation independent of a semantic model entirely. By leveraging an Instant-NGP [10] architecture, we introduce a procedure by which each sampled point computes a self-attention with a local field of sampled points around it, encouraging local consistency in the semantic field while maintaining the correct geometry. Specifically, we are able to stop the gradient flow from the semantic head into the density field, which allows for a higher accuracy geometry while giving way to a smooth and consistent semantic field without global context. This new semantic field can then

be *distilled* into the underlying semantic model to achieve generalization to novel scenes through a stronger 3D prior. In short, our key contributions are as follows:

- A generalizable method for transforming density/geometric embeddings into new semantic embeddings using a local self-attention module that preserves RGB performance and smooths the noisy semantic field.
- A fine-tuning approach that can generate 2D semantic segmentation supervision for out-of-distribution environments and lighting conditions.

2. Related Work

Neural Radiance Fields (NeRFs) are a class of implicit scene representations that can model a scene’s appearance, geometry [5], and in-general other spatially varying properties (such as semantics) [17]. The class of NeRFs varies significantly, with one of the major decisions being the trade-off between implicit and explicit representation: Specifically, traditional methods [9] that encode the entire scene as a coordinate based neural network are much more compact, but training time and fidelity falls as a result. On the other hand, more modern methods employ specific 3D structure such as voxel grids [7], spatial hashes [10], or general factors [1] and can be much faster for training and inference. Our work builds off of a framework called NeRFStudio [13] which employs a hybrid approach, using proposal networks and spatial hashing for fast and efficient training/inference.

2D Semantic Segmentation is a very heavily explored field, traditionally trying to solve the task of assigning pixels from a 2D image an object category from a distribution of categories for which the problem is defined. This can be seen as a generalization of the task assigning an arbitrary feature to a pixel, and as such, there has been a large rise in so-called “backbone-head” structures where a backbone such as ResNet [14] outputs a feature grid that is then interpreted by a final head to perform downstream tasks such as 2D semantic segmentation. Mask2Former [3] and DeepLabV3 [2] are two examples of these types of models, which we leverage in this paper. Nonetheless, because these 2D semantic segmentation models operate fundamentally in 2D, they are unable to capture 3D-aware information that might result from multi-view information that we have. As such, there is a need to “lift” the results from these models into 3D space.

NeRFs for 3D semantic segmentation. Lifting semantic segmentation using NeRFs was first proposed in [17], where ground truth segmentation labels were fused into an implicit scene representation for a variety of downstream tasks. However, this work did not consider the deep impact of inconsistent machine generated labels on the performance of this fusion, something we address in our proposed solution.

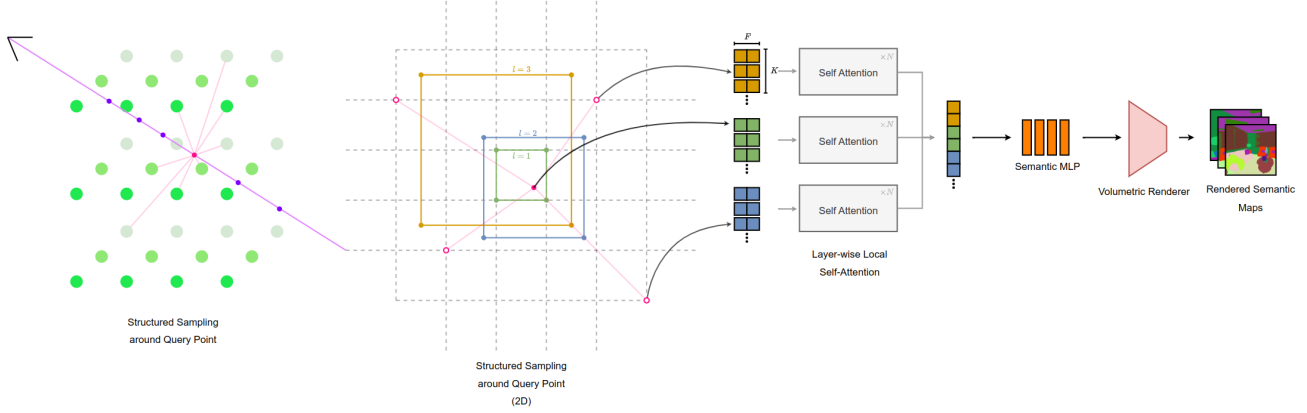


Figure 2. In AttentionNGP, each ray fired into the scene generates a set of query points, each of which results in a structured sampling of vertices of the underlying multi-resolution feature grid given in the Instant-NGP formulation. Where the features for the query points are computed through trilinear interpolation, the extra sampled vertices can query the hash-grid immediately, resulting in a set of features at each resolution for each query. From here, each set of features are forwarded through a layer-wise local self attention layer, and the final feature of the query point is assembled through the concatenation of the first entry in the attention layer’s output. This feature is then passed on to a semantic MLP that provides the bounded output needed to generate the output semantic maps through traditional volumetric rendering.

Since this work, several works have extended this line of study, either by adding panoptic lifting [11] or object fields [15]. In contrast to our solution, none of these methods propose the distillation of the lifted field into the model through fine-tuning, something we approach in our solution. Finally, continual adaptation [8] is one line of study that looks at doing this fine-tuning, but their lifting process requires a lengthy fine-tuning on ground truth labels, something we aim to avoid here.

3. Method

Given a set of posed RGB images $X_i = \{I_i, \mathbf{T}_i \in SE(3)\}$ we generate 2D segmentation maps $S = \{S_i\}$ using an out of the box segmentation model such as Mask2Former [3] or DeepLabV3 [2]. Combining these, we build an neural radiance field that jointly models appearance, density, and semantic output. Our approach lifts the 2D segmentation maps S as well as the images I into a 3D consistent implicit representation such that we add multi-view consistency on top of the machine generated semantic labels. Notably, we introduce a structured point based sampler and a series of localized attention blocks operating on density embeddings in order to encode local consistency and generalizability into the fitting process. From there, we then distill this multi-view consistency into the segmentation model through a fine-tuning process, enabling generalization of the original 2D segmentation model onto new scenes with superior 3D-aware performance.

3.1. Input Data

We can think of each input image I_i as a collection of rays $\{r_i^j \in R_i\}$ which we can transform into a global world frame using $\mathbf{T}_i$. Specifically, we parameterize each r^j (note that we subsequently drop the subscript i as our parameterization is invariant to the specific image) as $r^j = p^0 + td^j$ where p^0 is the ray origin (the camera origin expressed in world coordinates), d^j is the ray’s unit direction, and t is a free scalar.

For each ray $r_i^j \in R_i$, we may assign an RGB color $\hat{c}_i^j \in \mathbb{R}^3$ derived from the associated image I_i and a machine generated semantic class $\hat{s}_i^j$ derived from the segmentation map S_i . We note that we do not do any test time augmentation to improve the performance of the out of the box segmentation model like in other panoptic lifting approaches [6], instead relegating this task to our lifting process. We do, however, allow for $\hat{s}_i^j$ to be a probability distribution over semantic classes, which is important when the machine generated semantic labels are of low confidence and using the maximum would be detrimental to the lifting process. More details of this can be found in the experimental section.

3.2. Semantic Lifting

Attention motivation. Traditional NeRF models [8][17] that aim to model semantics usually independently treat each pixel, and therefore ray, during the forward pass of the rendering process. With two objectives, RGB and semantics, both predicted from the same embedding, there are significant conflicts between the observed density field, resulting in decreased performance depending on how the gradients are propagated back through the network. Because of this,

we commonly see one of two solutions: Either the semantic head is pulled out [6] and predicted directly from positional encoding, or RGB performance is degraded by passing on the semantic gradients [17]. However, both of these solutions are not desirable for generalizable scene understanding. The first method drops its scene-independent nature, and the second significantly reduces image quality, a key factor in fields like continual adaptation. Moreover, when the semantic gradients are stopped, additional supervision is required [8] (e.g. depth loss) or inconsistencies in the underlying semantic/feature model can significantly propagate into the implicit representation during training.

We aim to resolve this issue by leveraging the local consistency of an explicit grid-like structure in order to transform a collection of raw density embeddings into new semantic embeddings that are less noisy, more generalizable, and more amenable to multi-objective optimization. Specifically, we leverage the hybrid coarse-to-fine neural representation of Instant-NGP [10]. Recall that the Instant-NGP method places a multi-resolution feature grid with L levels (ranging from coarse-to-fine) into the scene, backed by a hash table used to preserve memory. Given a query ray r^j (note that we again drop the subscript i , as further discussion is independent of specific images), we sample N query points along the ray given by $\{q_n^j \in Q^j\}$. In order to compute the density σ and color c , we simply forward the point through the multi-resolution feature grid and associated MLPs as determined by the traditional Instant-NGP method.

Where our method differs from previous proposals, is in how we compute the semantics: for each query point q_n^j , we take the K_l closest vertices in each layer l and sample the associated feature embeddings by passing them through the hash-grid (an $O(1)$ operation) resulting in a “sequence” $D_l \in \mathbb{R}^{(K_l+1) \times F}$ in attention terms. Note that the 1st entry of this sequence comes directly from the trilinear-interpolation of the multi-resolution feature grid at the query point q_n^j . The sequence D_l is then passed through M layer-wise local self-attention layers in order to generate new semantic embeddings $S_l \in \mathbb{R}^{(K_l+1) \times F}$. The intuition behind this step is to think of the layer-wise attention as *clustering* similar density embeddings by assigning them higher weights in the associated attention matrix. After said attention step, the first row of each S_l , $S_{l,0}$ is taken as the query point’s semantic feature for that layer, and the full semantic embedding is given by $y = \bigoplus_l S_{l,0}$, or the concatenation of these rows. The final semantic distribution s is then computed by forwarding y through a small MLP. The full pipeline is given in Figure 2. Because Instant-NGP operates in a multi-resolution structure, one can think of each of the layer’s attentions blocks as learning a different clustering algorithm per resolution, akin to the hierarchical attention structure that dominates much of modern image processing. Furthermore, treating each sampled point independently is

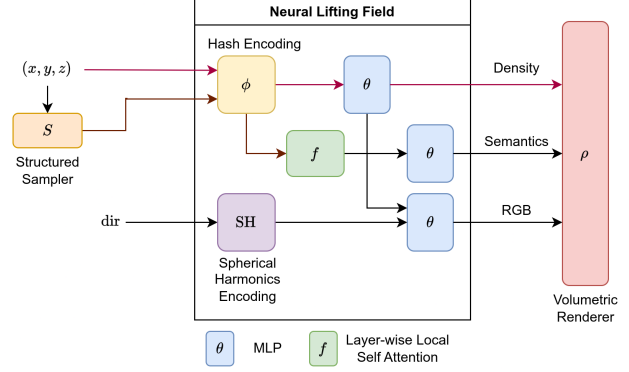


Figure 3. Neural Lifting field architecture. A 3D position and direction are fed into the network and passed through a series of encodings, MLPs, and our layer-wise attention to obtain the final outputs that are passed onto a volumetric renderer.

amenable to classical NeRF training techniques that involve computing the loss independently per-pixel, reducing the software complexity at the cost of performance.

Neural Lifting Field. We therefore extend the traditional NeRF field as can be seen in Figure 3. It is important to note that we stop the gradients between the hash encoding and layer-wise attention blocks. This decision allows us to obtain density and color values *exactly equivalent* to the traditional Instant-NGP formulation, theoretically guaranteeing no loss in RGB performance. Our final network therefore assigns to each 3D point a density $\sigma \in \mathbb{R}_{\geq 0}$, a semantic distribution of logits s , and an RGB color $c \in \mathbb{R}^3$.

Volumetric rendering. We omit the equations for the volumetric rendering of color, but we do include its extensions for semantic rendering. For each query point q_n^j (for a ray r^j), we first convert the associated semantic logits s_n^j to a distribution via the softmax function $s_n^{j'} = \text{softmax}(s_n^j)$. From there, we can apply numerical quadrature to approximate the volumetric rendering integral:

$$s(r^j) = \sum_{n=1}^N T_n^j \alpha_n^j s_n^{j'} \quad (1)$$

where $\alpha_n = 1 - e^{-\sigma_n^j \delta_n^j}$, $T_n^j = \prod_{k=1}^{n-1} (1 - \alpha_k^j)$ with δ_n^j being the distance between adjacent query points along the ray.

Distillation From here, we can re-render each image and semantic map from the corresponding poses $\mathbf{T}_i$ to form NeRF generated images $\tilde{I} = \{\tilde{I}_i\}$, $\tilde{S} = \{\tilde{S}_i\}$, where we can then fine tune our out-of-the-box segmentation model using the traditional cross-entropy loss for training semantic

segmentation models, the goal being to move the 3D-aware knowledge from the NeRF model into the 2D semantic segmentation model.

3.3. Loss Functions

Appearance Loss. Because we process RGB rays independently (not as patches), our appearance loss is the mean squared error between the rendered image and the ground-truth image:

$$L_{\text{RGB}}(R_i) := \frac{1}{|R_i|} \sum_{r_i^j \in R_i} \|c_i^j - \hat{c}_i^j\|^2 \quad (2)$$

Semantic Loss. Here, we do employ patch processing, and as such, our semantic loss is defined as the cross entropy of the obtained semantic distribution labels relative to the machine generated semantic labels:

$$L_{\text{sem}}(P_k) := \frac{1}{|P_k|} \sum_{r^j \in P_k} \sum_{s_i \in s} \hat{s}_i^j \log(s_i^j) \quad (3)$$

Together, we optimize the combined loss $L := \lambda_{\text{RGB}} L_{\text{RGB}} + \lambda_{\text{sem}} L_{\text{sem}}$

Implementation Details. We heavily build upon the nerf-studio [13] framework in this work and specifically the nerfacto method, which is implemented with Pytorch. We note that nerfacto is built upon Instant-NGP, the only difference being that nerfacto uses a set of proposal samplers to generate appropriate samples along rays. We trained our model using the Adam optimizer with a learning rate of 1×10^{-2} annealed to 1×10^{-4} annealed over 20k iterations. The entire training takes 60K iterations and is trained on a variety of GPUs (slurm cluster).

4. Datasets

Replica Replica [12] is a 3D dataset comprised of high-fidelity indoor room-based scenes and the capability to render dense geometry, appearance, and semantic information from arbitrary trajectories. We use the pre-rendered data made available from SemanticNeRF [17] which consists of randomly roll-locked trajectories to render a series of 900 640x480 images with a default pin-hole camera model that we map to the SDFStudio [16] camera model. The Replica dataset consists of 88 semantic classes which we manually map either to the 150 classes of the Mask2Former [3] pre-trained model corresponding to a subset of the ADE20k [18] [19] semantic classes or the 21 classes of the DeepLabV3 [2] model. We additionally take pre-rendered Replica data from MonoSDF [16] and apply the same semantic class mapping as above.

ScanNet ScanNet [4] is another 3D indoor dataset with 2.5M views in 1513 scenes including semantic segmentation, camera poses, and surface reconstruction. We take from this the pre-rendered images, semantic segmentation labels, camera poses and optionally depth and sample down the roughly 700 1296x968 images to 150 384x384 images which we feed into our pipeline. We additionally map the semantic labels from the NYUv2-40 definition and map them to the Mask2Former or DeepLabV3 semantic classes as we do above.

5. Experiments

We will evaluate the effectiveness of our semantic lifting on the tasks of novel view semantic segmentation and novel view synthesis. The effectiveness of our distillation process will be evaluated through novel view semantic segmentation as well as well as multiview accuracy from ground-truth lifted segmentation data.

Metrics. For quantitative evaluation, we are mainly looking at image and semantic reconstruction accuracy. Image reconstruction accuracy is calculated primarily through the peak signal to noise ratio (PSNR). Semantic reconstruction accuracy is evaluated through accuracy (ACC), mean average precision (mAP), and the mean intersection over union (mIOU), which is calculated through the Jaccard Index.

5.1. Results

We compare with state of the art Panoptic Lifting [6] by taking the semantic results and image reconstruction accuracy from their reported results. We additionally report fine-tuning data for our underlying semantic models.

As shown in Table 1 we successfully are able to lift semantic understanding into a multiview consistent space and distill it back into the underlying semantic models. While we do not outperform the state of the art, we come within a few percentage points in the mIOU metric but also significantly increase the image reconstruction accuracy at the same time. This signifies that our model is more balanced, able to perform well on a variety of tasks, many of which are multi objective. Panoptic lifting seems to suffer in terms of image reconstruction accuracy mostly due to its architecture, and our performance might be a result of model capacity, as it still remains to be seen.

Qualitatively (see 4), we see that Mask2Former [3] struggles on the Scan-Net dataset while DeepLabV3[2] struggles with the Replica dataset. However, after lifting and distillation, we are able to significantly increase performance on these out of distribution examples. Additionally, we can more clearly see the difference the attention layers make as can be seen by the significant noise without these layers. This will be expanded on in the ablations section.

Method	Replica[12]				ScanNet[4]			
	mIOU↑	ACC↑	mAP↑	PSNR↑	mIOU↑	ACC↑	mAP↑	PSNR↑
Mask2Former[3]	52.4	56.8	64.6	-	10.2	14.4	15.6	-
DeepLabv3[2]	28.4	30.2	32.9	-	60.2	63.6	68.1	-
Panoptic Lifting[6]	67.2	-	-	28.1	65.2	-	-	28.5
AttentionNGP (Ours)	65.6	68.8	72.2	38.1	64.7	69.2	70.1	30.1
Mask2Former (Fine-Tuned)	64.4	65.6	69.2	-	64.5	68.2	69.9	-
DeepLabv3 (Fine-Tuned)	63.2	65.1	69.2	-	62.1	64.4	69.3	-

Table 1. Quantitative evaluation on novel views from the eval set. We do not outperform Panoptic Lifting when it comes to lifting performance, but we note that panoptic lifting chooses to directly model the semantics as a function of position, whereas we try to model semantics as a function of a local feature embedding neighborhood around a query point. Additionally, Panoptic Lifting does not report mAP or ACC, so we can’t evaluate it concerning these two metrics. However, when taking into account the combined image reconstruction accuracy and semantic performance, we do outperform Panoptic lifting.

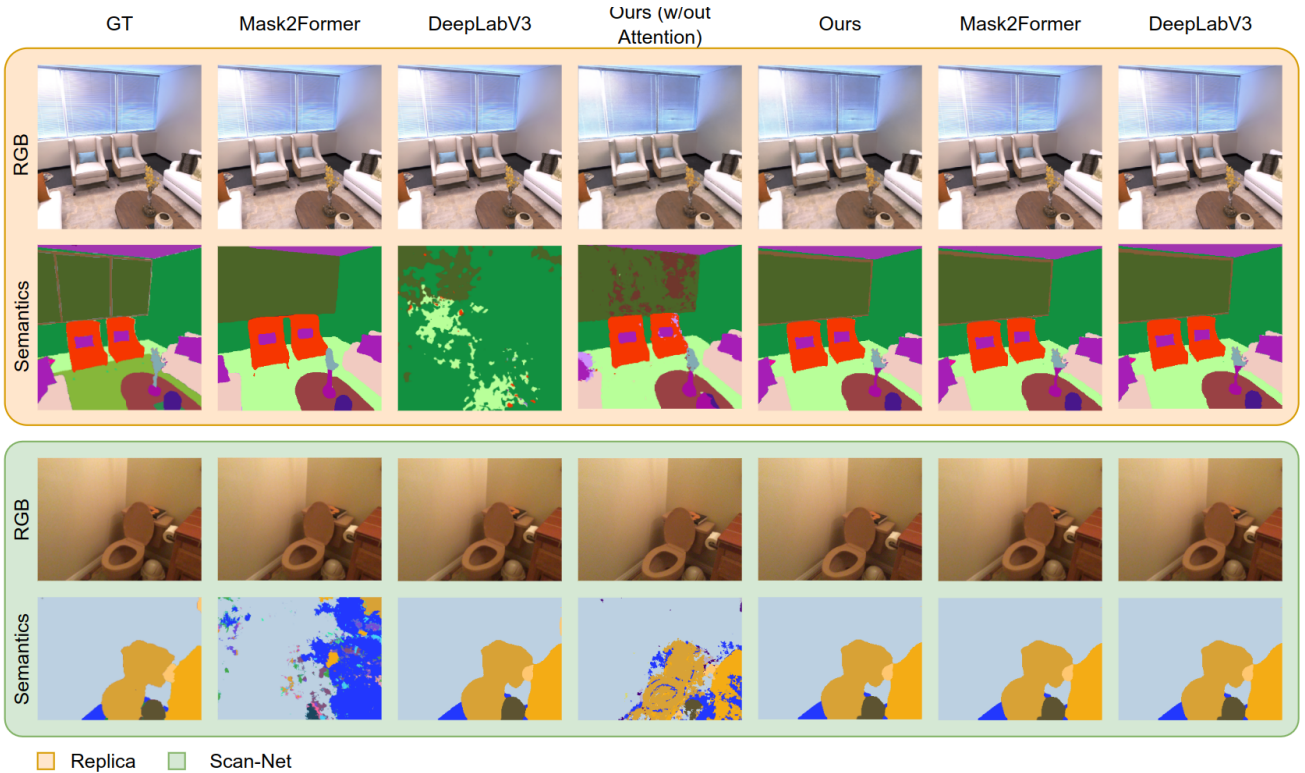


Figure 4. Qualitative evaluation on novel views from the eval set. We see that our full model correctly matches the ground truth RGB and semantics while our model without the attention layers fails to correctly get de-noised semantics. On our fine-tuning, we see that we can successfully distill 3d aware knowledge into the underlying semantic model.

5.1.1 Ablations

In Table 2, we show 3 different models we tested using the Mask2Former semantic model on the Replica Dataset.

Attention. When dropping the attention portion of the model, we are still treating the RGB and color the same, meaning that while the PSNR should remain the same, the

semantic accuracy should decrease significantly. This can easily be seen by observing the large drop in mIOU in Table 2, row 1. This is equivalent to using a feature embedding that has been optimized to match the RGB outputs as a semantic embedding, and you can clearly see the noise as observed in Fig 4. Additionally, the PSNR is marginally higher because of the multi objective formulation of Nerfacto and the interesting dynamics between the interlevel losses of the proposal

Ablation Type	mIOU↑	ACC↑	mAP↑	PSNR↑
Ours (w/out Attention)	24.4	26.2	28.4	40.1
Ours (w/out Gradient Block)	66.7	70.2	73.4	22.9
Ours	65.6	68.8	72.2	38.1

Table 2. Ablations using Mask2Former as the semantic model on the Replica dataset. The model we choose achieves a balanced proportion of accuracy towards the semantic labels and the RGB image reconstruction accuracy.

sampler.

Gradient Blocking. This model keeps the attention portion and allows the gradients from the layer-wise attention to feed back into the hashgrid, thereby being able to influence the density and RGB values later in the pipeline. Because the semantic portion of the model can now be end-to-end optimized, we see a marginal increase in performance in terms of semantics (Table 2, row 2) but a significant dip in image reconstruction accuracy primarily due to inconsistency in the underlying density field. The model we choose has a good mix of accuracy in terms of semantics and image reconstruction accuracy, which is the two metrics we are aiming to maximize. Empirically, we observe that the performance of the semantic branch becomes proportional to the number of attention layers, which signifies an unbound capacity that we did not have time to maximize as well.

5.1.2 Failures and limitations

I did attempt to freeze the attention layers and use them with a different semantic model and scene, but this failed to come to fruition after many attempts. I expand more upon this in the video presentation, but there is still a clear gap between the generalized semantic understanding that should be the goal. More concretely, it would be interesting to explore things like open-set segmentation that relies even more directly on this attention-like clustering approach to “discover” semantic classes instead of learning them from an underlying model.

6. Conclusion

We have introduced AttentionNGP, an approach that uses a structured sampler and a series of layer-wise local attention layers to transform the explicit density embeddings initialized in the scene through the multi-resolution feature grid of Instant-NGP into new semantic embeddings that have a local consistency that prevents noise from the underlying machine semantic models from being lifted into a multiview consistent 3D space. Furthermore, we are able to distill this 3D

aware knowledge back into these semantic models through a fine-tuning process, significantly improving their performance on out-of-distribution data. I believe that pursuing this idea of a graph on the Instant-NGP feature embeddings can be a significant step forward in 3D scene understanding.

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